

Theorem (Fundamental Theorem of Finite Abelian Groups).

Exercise 1. Find 5 different Abelian groups of order 16.

Exercise 2. How many Abelian groups are there of order 36 (up to isomorphism)?

Exercise 3. For what values of n will there be exactly 4 isomorphism classes of groups of order n ?

Exercise 4. Find all Abelian groups, up to isomorphism, of order 360.

Exercise 5. Show that there are two Abelian groups of order 108 that have exactly 13 subgroups of order 3.

Exercise 6. The set $G = \{1, 4, 11, 14, 16, 19, 26, 29, 31, 34, 41, 44\}$ is a group under multiplication modulo 45. Write G as an external and an internal direct product of cyclic groups of prime power order.

Exercise 7. Suppose the order of some finite Abelian group is divisible by 10. Prove that the group has a cyclic subgroup of order 10.

Exercise 8. Let G be an Abelian group of order 16. Suppose that there are elements a and b in G such that $|a| = |b| = 4$ and $a^2 \neq b^2$. Determine the isomorphism class of G .